SOURABH PORWAL

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Functional safety engineer with seven years in safety-critical systems. Currently own functional safety at Vay, whose remotely operated fleet of ~200 vehicles runs on public roads in Las Vegas with no driver on board. With no regulation in place for teledriving, helped build the in-house safety function from scratch and set up its safety framework. Lead system-level hazard analysis (HARA, FMEA, FTA), risk assessment, and safety cases, and now lead the safety work as Vay brings AI into the driving task itself. Hands-on across the full ISO 26262 lifecycle and experienced in leading external safety assessments, with a systems-safety foundation that carries directly into robotics and other emerging, autonomous platforms.

CORE COMPETENCIES

Safety Engineering: System-level hazard analysis & risk assessment (HARA, FMEA, FTA); safety cases; traceable safety requirements, V&V & certification audit support; ISO 26262, ISO 21448; familiarity with ISO 13849 and IEC 61508 principles via automotive functional safety

Safety Strategy & Regulatory: Building a safety function from scratch; adapting safety processes for emerging, autonomous technologies; regulatory monitoring & compliance; independent safety assessments (TÜV SÜD)

Systems & AI: Cross-functional safety across hardware, software & systems; embedded systems & sensor characterisation; functional safety of AI-based systems; LLMs (general working knowledge)

EXPERIENCE

Senior Functional Safety Engineer

Vay Technologies GmbH · Berlin, Germany · 09/2023 – Present

- Own functional safety for the remote-driving system, leading ISO 26262 compliance, partnering with engineering, leadership, and legal to embed safety from concept through deployment.
- Lead the safety work for bringing AI into safety-critical driving functions.
- Own and author the safety case as per UL 4600, and define the functional and technical safety architecture, including system level redundancies, for a remote-driving fleet of ~200 vehicles now operating on public roads in Las Vegas with no driver on board.
- Derive tolerable failure-rate thresholds for each fallback-manoeuvre type from probabilistic risk analysis, with an operational stop-fleet trigger if thresholds are exceeded.
- Work with the embedded software team on the functional safety of the ASIL D microcontroller platform (Infineon TriCore), and characterise safety relevant sensors.
- Define the interface between the safety controller and the vehicle over CAN and LIN; analyse failure modes across the system, and specify the corresponding system reactions.
- Act as safety lead for integrating the remote-driving stack into a B2B customer's autonomous driving product as a fallback layer, working directly with the partner's safety and engineering teams on architecture, integration, and safety deliveries.

Functional Safety Engineer II

Vay Technologies GmbH · Berlin, Germany · 06/2022 – 08/2023

- Designed the system's fail-safe fallback behaviour (Minimum Risk Manoeuvre) end-to-end, from concept and systems through functional safety, incorporating regulatory and legal considerations.
- Led the functional-safety side of the independent TÜV SÜD assessment of the system's safety concept and risk analyses, contributing to the endorsement that enabled Vay's first-in-Europe approval to operate on public roads without a safety driver.
- Performed safety analyses (FMEA, FTA) and supported verification and validation at system and vehicle level.

Functional Safety Engineer I

Vay Technologies GmbH · Berlin, Germany · 06/2021 – 06/2022

- Helped establish Vay's in-house functional-safety function and processes from scratch, deriving the safety framework from first principles in the absence of any teledriving regulation.
- Implemented requirements management and traceability (JAMA).
- Contributed to the company's first successful external safety assessment (TÜV SÜD).

Functional Safety Engineer / Manager

BERNS Engineers GmbH · Gilching, Germany · 01/2020 – 06/2021

- Prepared safety concepts and analyses (FSC, TSC, HARA, FTA) for OEM customers and worked directly with their architects and engineers on ASIL D ADAS features.
- Served as functional safety manager on certain projects, owning the safety plan and overseeing supplier safety activities.

Functional Safety Engineer

AKKA Technologies · Ingolstadt, Germany · 06/2019 – 12/2019

- Supported ISO 26262 development for an ASIL B system and managed vehicle configuration data.

Junior Executive

Fleetguard Filters Pvt. Ltd. · Pune, India · 08/2013 – 09/2015

- Managed production part approval process (PPAP) activities across manufacturing suppliers, including process FMEA and process flow documentation; reduced component cost by ~25% through supplier negotiation.

EDUCATION

M.Sc. Mechanical Engineering

Rhine-Waal University of Applied Sciences · Kleve, Germany · 10/2015 – 04/2019

B.E. Mechanical Engineering

Visvesvaraya Technological University · Belgaum, India · 08/2009 – 04/2013

CERTIFICATION

CertX – ISO 26262:2018 Functional Safety Red Belt (03/2021)

PERSONAL DETAILS

Nationality: German
Mobility: Open to relocation
Languages: English (fluent), German (B2), Hindi (native)